

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall enforce dose interval rules (e.g., minimum 28 days after Dose 1) before unlocking Dose 2 booking for a citizen.	High	Pass: Dose 2 booking blocked if interval < 28 days; Fail: Early vaccination slot confirmed.	Medical compliance.
FR-002	Functional	The system shall allow a Citizen Registrant to register using basic demographic details.	High	Pass: Profile created successfully; Fail: Missing mandatory fields trigger an error.	Establishes user identity.
FR-003	Functional	The system shall permit a Citizen Registrant to schedule a vaccination slot at an available center.	High	Pass: Slot reserved and capacity reduced by 1.	Core scheduling function.
FR-004	Functional	The system shall enable the Vaccination Officer to mark a scheduled dose as administered.	High	Pass: Status changes to 'Administered'.	Required for tracking.
FR-005	Functional	The system shall generate a verifiable QR digital vaccination certificate.	High	Pass: QR code generated upon dose completion.	Proof of vaccination.
NFR-001	Nonfunc.	The vaccination certificate verification endpoint shall authenticate digitally signed QR codes offline/online in under 150 ms.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Performance & Security.
NFR-002	Nonfunc.	The system shall support at least 5,000 concurrent users during rollout announcements.	Medium	Pass: No degradation in response time under a 5,000-user load test.	System reliability.